

1. Process

1.1. Establish Randomisation and Container Codes Strategy

Randomisation and Container Code Strategy and planning is required and documented prior to the generation of randomisation and container codes. The timing of the development of the RCCS varies from study to study but is typically performed when there is a near-final protocol in place and in line with logistic operational study planning.

The content of the RCCS supports the intended blind strategy and reflects the randomisation requirements for treatment assignment, container-number labelling and packaging needs as well as describing access requirements for randomisation and container codes prior to protocol-defined reporting milestones.

A standard RCCS template is used to ensure all components are addressed.

For specific situations where the RCCS contains sensitive information access should be restricted to only those who need it. The RCCS, in these situations, is not to be stored in TMF until after the study has been formally un-blinded.

e.g.

- Multi-period crossover design with sentinel subjects used at each dosing period
- Dose-finding study where a dose is added or dropped as part of an instream decision and the decision is to be concealed from the broader study team.

The following table depicts the roles and responsibilities for creation and approval of the Randomisation and Container Codes Strategy (RCCS).

Step	Process Step	Role
1.	Develops the draft RCCS using the template. The Access Plan section of the RCCS identifies those individuals needing access to the codes who do not have direct access through RandomAll NG interface (see footnote and Section 6.3). Where the RCCS contains sensitive information, the sensitive information is to be redacted prior to broader key contributor review. Otherwise, there is no restriction on method of access for review.	Study Statistician (and SCSL as applicable)
2.	Submit RCCS to key contributors for review.	Study Statistician
3.	Review the RCCS to ensure the approach meets the requirements for the study: SAP: to ensure the RCCS meets the study design requirements as well as study execution logistics SCSL (if applicable): to ensure packaging/labelling approach supports the blind strategy IRT Representative (if applicable): to ensure RCCS is compatible and aligned with IRT capabilities Other Key Contributor(s) (if applicable): • to be added as needed to support complex study designs (e.g. Study Programmer, Clinical Pharmacologist, Operations Study Lead)	Study Statistician, SAP, and where applicable (SCSL, IRT Representative and/or Others)

Step	Process Step	Role
	If RCCS passes review, go to next step. If RCCS does not pass review, then resolve and repeat step 2 onward	
4.	Reviews the RCCS to ensure that only permitted roles are identified as planned recipients of randomisation schedules or container lists prior to full code release. Permitted roles will be limited in accordance with the level of blinding specified in the protocol and, if applicable, an IDMC or Data Review Committee Charter.	Statistics Line Manager
5.	Obtains approvals as applicable from Statistics Line Manager, SAP, SCSL IRT Representative and/or other key contributor(s) - Where the RCCS contains sensitive information, the approvals are by email approval. - Otherwise the approvals are an electronic signature within the Trial Master File (TMF) system (or if this is not possible an email approval).	Study Statistician
6.	Where information within the RCCS is not sensitive, ensure the RCCS and approvals are final in the TMF (refer to Appendix 2).	Study Statistician
7.	Where information within the RCCS is sensitive: Statistician sends RCCS to Randomisation Coordinator mailbox Statistician files the email approvals in TMF and ensures a cover page for the RCCS is in place (with version, date, and location of approved RCCS). Randomisation Coordinator files the RCCS in a secure place following appropriate steps outlined within Randomisation Coordinator Guidance.	Study Statistician and Randomisation Coordinator
8.	Where an amendment to the RCCS is required, determines if a Randomisation Change Control Plan is required (Section 6.2.4) then repeats steps 2-4. Note: an RCCS amendment is needed if - protocol amendment requires a change to the agreed strategy - new randomisation schedule needed - new container list needed - additional access to the randomisation or container list is required	Study Statistician

Note: Where the protocol stipulates an Independent Data Monitoring Committee (IDMC) is to be convened and an independent Statistical Data Analysis Center (SDAC) is contracted to produce interim unblinded outputs, the SDAC must also be solely responsible for creating unblinded datasets. In such circumstances, ABC statisticians and programmers (including SDTM programmers and Functional Service Provider staff who are ABC account holders) are prohibited from being designated as recipients of randomisation codes in the RCCS for SDAC-delivered interims.

1.2. Generate Randomisation and Container Codes

Code generation is a two-stage process whereby randomisation and container codes are drafted for review/approval and then finalised. Unless described as draft, the terms

“randomisation codes” and “container codes” refer to the actual codes implemented in the study for treatment/intervention assignment and container labelling/dispensing respectively.

In accordance with the trial design, one or multiple randomisation schedules and container lists are generated as required. For example, if it is planned to analyse a multi-cohort study upon completion of each cohort, separate lists are created for each cohort. This allows independent release of milestone-specific randomisation schedules and container lists (Section 4).

1.2.1. Generate Randomisation Codes Using RandomAll Next Generation (NG)

Where RandomAll NG capabilities can support the study design, randomisation codes are generated using this ABC validated system. If the trial design is not supported by RandomAll NG, then single-use-software is permitted (refer to Section 6.2.2).

Step	Process Step	Role
1.	Create draft randomisation codes in accordance with the most recent approved version of the RCCS	Study Statistician
2.	Review the draft randomisation codes and approve if they meet the criteria defined in the RCCS. If rejected, then return randomisation codes to draft status and make corrections. Repeat the review and approve when they meet the criteria defined in the RCCS.	Study Statistician and QC Statistician
3.	Publish the approved draft randomisation codes.	Study Statistician
4.	When an IRT is used, the Study Statistician waits for outcome from IRT test set up before generating final randomisation codes. • If IRT test set up is not successful and results in a required change to the randomisation codes, then amend the RCCS (follow steps in Section 6.1) and re-create draft randomisation codes (from step 1 above). • If IRT test set up is successful or if there is no IRT used, continue to next step.	Study Statistician
5.	Finalize the randomisation codes and notify study team. ① RandomAll NG system conceals the actual randomisation codes from Study Statistician until release of randomisation codes (Section 6.4).	Study Statistician
6.	When additional randomisation codes are required and there is no change to the treatment allocation ratio or stratification, then amend the RCCS (follow steps in Section 6.1) to reflect a new randomisation number range and append the randomisation schedule in RandomAll NG. Otherwise, a new randomisation schedule is needed (refer to Section 6.2.4)	Study Statistician

1.2.2. Generate Randomisation Codes Using Single-Use Software

Where the randomisation strategy cannot be supported by RandomAll NG, randomisation codes are generated by single-use software (e.g., SAS code). Exceptional care is needed to maintain concealment of the randomisation codes and check quality/accuracy of creation method. A Randomisation Coordinator is enlisted to support the generation of the randomisation codes.

Where randomisation codes are generated in advance and ABC systems are used for management of treatment assignment, the following steps are followed.

Step	Process Step	Role
1.	Create the Randomisation QC Record using the template. Identify and assign Programmer, QC Statistician and Randomisation Coordinator responsibilities for the generation of draft and final randomisation codes.	Study Statistician
2.	Create software to generate draft randomisation codes using the specifications within the RCCS. Where a seed is used to generate the draft randomisation codes, select a seed. Generate draft randomisation codes. Provide access to randomisation code documentation to QC Statistician: [This includes (as applicable) RCCS, software used to generate draft randomisation codes, seed selected, block size(s) used to generate the draft randomisation schedule(s) and Randomisation Codes QC Record.]	Study Programmer or Statistician
3.	Verify the software accurately generated draft randomisation schedules that meet the specifications outlined in the RCCS: • correct schedule numbers • correct treatments group definitions • correct period specific treatments • correct stratification groups • correct randomisation number ranges • correct treatment allocation ratio Verify the software ran with no errors or warnings. Record outcome in the Randomisation Codes QC Record.	QC Statistician
4.	Where the QC failed, resolve all issues and repeat steps 2 and 3 until successful QC achieved.	Study Programmer or Statistician
5.	Record outcome in the Randomisation QC Record and provide to Study Statistician.	QC Statistician
6.	Define the treatments and treatment groups in RandomAll NG aligned with those defined in the verified software.	Study Statistician

Step	Process Step	Role
	Provide access to RCCS, verified software, seed selected for generation of draft randomisation codes (where applicable), draft randomisation codes and Randomisation QC Record to Randomisation Coordinator	
7.	The following work is conducted in a secured area restricted to Randomisation Coordinators: • if a seed is used, select seed that is unique from that used to generate the draft randomisation codes • generate final randomisation codes using provided software and new seed	Randomisation Coordinator
8.	Perform QC of final randomisation codes comparing the details of the draft randomisation codes to details of the final randomisation codes. Record outcome in Randomisation QC Record. Where QC fails, resolve all issues with Randomisation Coordinator and repeat step 7 and 8 until successful QC achieved.	QC Randomisation Coordinator
9.	Import draft randomisation codes into RandomAll NG database following process detailed in Randomisation Coordinator Guidance. Notify Study Statistician when complete.	Randomisation Coordinator
10.	Perform a QC of imported draft randomisation codes by comparing draft randomisation codes in RandomAll NG with draft randomisation codes approved in step 4 and 5. Record outcome in Randomisation QC Record. Where QC fails, resolve all issues with Randomisation Coordinator and repeat step 9 and 10 until successful QC achieved. Notify Randomisation Coordinator when complete.	Study Statistician
11.	Import final randomisation codes into RandomAll NG.	Randomisation Coordinator
12.	Perform a QC of imported final randomisation codes by comparing final randomisation codes in RandomAll NG with final randomisation codes approved in step 8. Record outcome in the Randomisation Codes QC Record. Where QC fails, resolve all issues with Randomisation Coordinator and repeat step 11 and 12 until successful QC achieved. Notify Study Statistician when complete. Maintain programming related files for generation of the final randomisation codes.	QC Randomisation Coordinator
13.	Notify study team final randomisation codes are generated and available in the RandomAll NG randomisation database File Randomisation Codes QC Record in TMF (refer to Appendix 2).	Study Statistician

Step	Process Step	Role
	① RandomAll NG system conceals the actual randomisation codes from Study Statistician until release of randomisation codes (Section 6.4).	
14.	Where additional randomisation codes are required and there is no change to the treatment allocation ratio or stratification, amend the RCCS (follow steps in Section 6.1) to reflect a new randomisation number range and modify the verified software to reflect the new randomisation number range and repeat from step 3 above. Otherwise, a new randomisation schedule is needed (refer to Section 6.2.4).	Study Statistician

1.2.3. Full Release of Codes and Creation of Associated Unblinded Datasets

The steps below are followed when protocol-defined milestones require full release of randomisation codes (e.g. end-of-study reporting or reporting of a completed cohort). Once released, the codes are accessible for data unblinding, reporting and data reconciliation capabilities. Unused codes that have been released should not be used for further enrolment.

Step	Action	Role
15.	Create a Randomisation Release Request (RRR) using the standard template and submit to the Statistics Line Manager. - indicate randomisation schedule(s) used for treatment assignment and container list(s) for dispensing (where appropriate), - provide supporting documents i.e. Data Manager database milestone notification	Study Statistician
16.	Review RRR: - confirm database milestone is achieved for protocol-specific milestone - for a stand-alone cohort, confirm correct randomisation schedule(s) and container list(s) are requested for release by comparing the details provided in RRR with the RCCS If approve the request - release (unblind) the randomisation schedule(s) and container list(s) from RandomAll NG - enter date of release on RRR and return to Study Statistician If request is not approved notify statistician with reason for denial	Statistics Line Management
17.	If request is approved and ABC is responsible for preparing the SDTM datasets, then notify Study SDTM Unblinding Programmer the final codes are available for unblinding If request is approved and third party is responsible for delivery of blinded SDTM datasets, then notify the ABC Clinical Programmer the final codes are available for unblinding	Study Statistician

Step	Action	Role
	If request is approved and a third-party provider is responsible for preparing the unblinded SDTM datasets follow the distribution process as outlined in Section 3.1	
18.	If ABC is responsible for preparing the SDTM datasets • associate treatment details (i.e. randomisation codes, container codes and/or any treatment deviations identified in Section 6.5) with clinical trial data. • notify Study Statistician, Study Programmer, SDTM Delivery Lead and Data Manager when complete.	SDTM Unblinding Programmer
19.	If blinded SDTM datasets are delivered by a third party for ABC analysis and reporting, then • associate treatment details (i.e. randomisation codes, container codes and/or any treatment deviations identified in Section 6.5) with clinical trial data. • notify Study Statistician and Data Manager when complete.	Clinical Programmer
20.	For a stand-alone cohort that concludes prior to completion of study, send notification to SAP, SCSL and IRT Representative that the specific released randomisation schedule(s) and container list(s) are retired (i.e. should not be used for future treatment assignment and treatment dispensing).	Study Statistician
21.	For randomised studies where the treatment/ intervention assignments are concealed from the sponsor (e.g., sponsor blind, double blind) and the protocol-defined reporting milestone requires treatment reconciliation of safety data Reconciliation of Adverse Event, Pregnancy and Incident Reports), send notification to Central Safety mailbox within two business days.	Study Statistician
22.	For studies where the treatment/intervention assignments are concealed from the investigator (e.g., double-blind), prepare site specific list of treatment/intervention assignments for each randomised subject after completion of full statistical analysis for final protocol-defined reporting milestone in support of POL_87187 Policy on Human Subject Research.	Study Statistician or Study Programmer
23.	Once analysis for final protocol-defined reporting milestone is completed, • create a copy of the RandomAll NG audit trail • check that relevant TMF documentation previously filed by Randomisation Coordinator is accessible (i.e. TMF blind is lifted) if applicable (Sections 6.1 and 6.3.1) • request sensitive programming support files from Randomisation Coordinator if applicable • file RandomAll NG audit trial and any sensitive programming support files into TMF	Study Statistician

File for Trial Master File

Documents that are generated during this process and that do not jeopardise concealment of the randomisation are filed in the study file and managed in accordance with local study sponsor record management procedures.

Following release of the randomisation at the end of the study and completion of statistical reporting or where activities are in relation to study termination, **Study Statistician** obtains any sensitive documentation being maintained by Randomisation Coordinator (e.g., distributions of randomisation codes, creation of randomisation codes) and files the documentation in the study file.

Trial Master File content and responsible role related to required documentation created in concert with this procedure is summarised in the table below:

Document	Responsible Role
Randomisation and Container Codes Strategy (RCCS) Document and approvals	Study Statistician
Randomisation codes (included as a listing for the Clinical Study Report). The schedule remains in RandomAll NG.	Study Statistician
Evidence of creation and release of Randomisation and Container codes - RandomAll NG Audit Trail or legacy RandomAll Audit Trail (if applicable) - request for Release of Randomisation and Container Codes - programming and Randomisation QC Record for creation of randomisation codes using single use software (if applicable)	Study Statistician
Approval of Access for non-routine reason from VP and Head Biostatistics (if applicable)	Study Statistician
Requests for and approval of distribution of randomisation and container codes (if applicable)	Study Statistician
Distribution of randomisation and container codes documentation (where applicable)	Study Statistician
Randomisation change control plan (if applicable)	Study Statistician